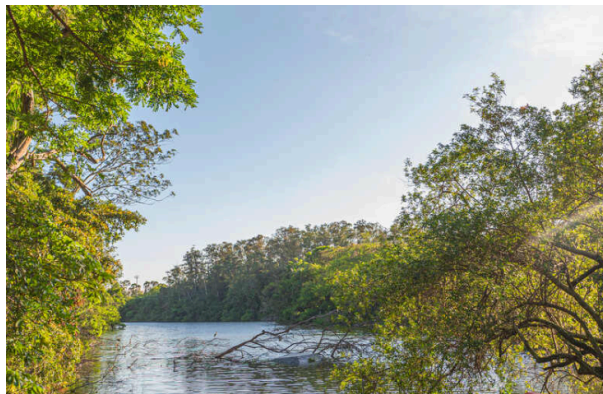


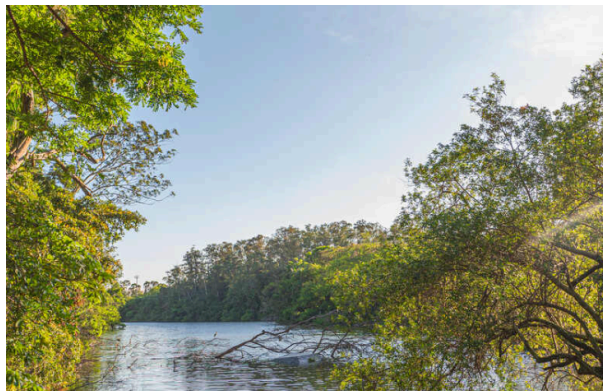
Support, Protection, and Movement

Integument and Its Functions

- **Integument** is the outer covering of the body, providing **protection**, **mechanical support**, and **barrier** against bacteria and fluid loss.
- In **invertebrates**, the principal covering is a single-layered **epidermis**, sometimes with a secreted **cuticle** for extra protection.
- **Arthropod** integument consists of a single-layered epidermis (hypodermis) that secretes a complex cuticle of two zones: **procuticle** (protein and chitin) and **epicuticle** (proteins and lipids).
- **Calcification** (crustaceans) and **sclerotization** (insects) harden the cuticle.
- In **vertebrates**, the integument includes a thin, outer **epidermis** (from ectoderm) and a thicker, inner **dermis** (from mesoderm).
- The **epidermis** gives rise to derivatives such as **hair**, **feathers**, **claws**, and **hooves** through **keratinization**.
- The **dermis** contains blood vessels, nerves, pigment cells, and supports the epidermis.
- **Keratin** provides strength and flexibility to structures like scales, feathers, and hair.
- The **scaling law** explains that as animal size increases, **weight** increases with the cube of length ($\text{weight} \propto \text{length}^3$), but **bone strength** increases with the square of length ($\text{strength} \propto \text{length}^2$). This means larger animals need disproportionately thicker bones.



- **Fish scales** are bony dermal structures; **reptilian scales** are keratinized epidermal structures.

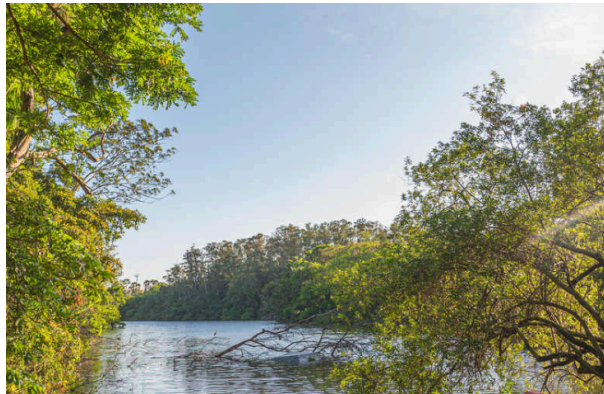


- **Claws**, **beaks**, and **horns** are combinations of epidermal (keratinized) and dermal components.



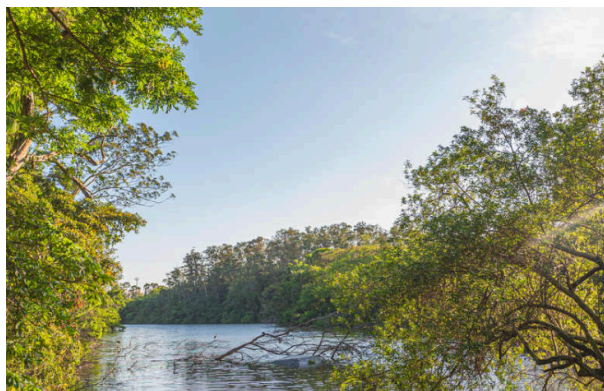
Animal Coloration

- **Coloration** is produced by **pigments** (biochromes) or by **structural color** (physical structure reflecting light).
- **Chromatophores** are pigment-containing cells; **melanins** produce black/brown, **carotenoids** produce yellow/red, and **iridophores** create metallic effects.



Skeletal Systems

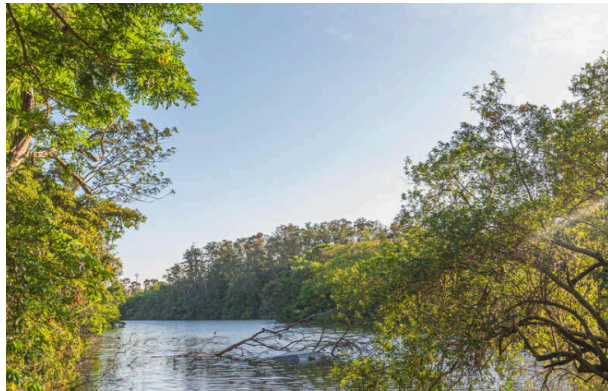
- **Skeletons** provide **support**, **muscle attachment**, and **protection** for organs.
- **Hydrostatic skeletons** use body fluids for support (e.g., earthworms).



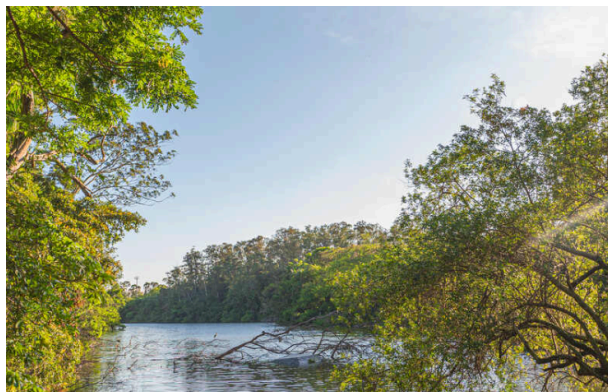
- **Muscular hydrostats** (e.g., elephant trunk) use muscle bundles for movement without skeletal support.



- **Rigid skeletons** include **exoskeletons** (invertebrates) and **endoskeletons** (vertebrates).
- **Bone** is a living tissue with inorganic calcium salts and collagen fibers, providing strength and flexibility.
- **Compact bone** and **spongy bone** are two types of bone structure.



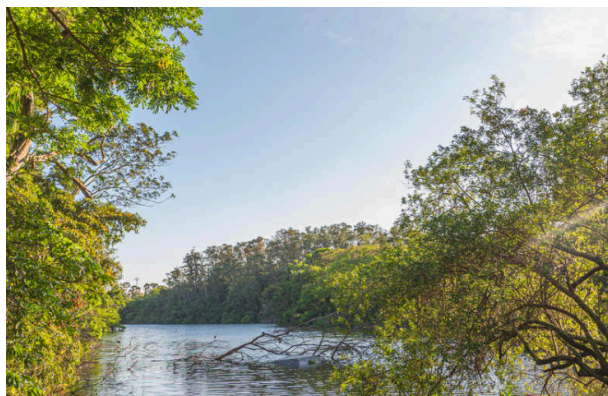
- The **vertebrate skeleton** is divided into **axial** (skull, vertebral column, ribs) and **appendicular** (limbs, girdles) skeletons.



- The **human skeleton** shows both primitive and specialized features, with adaptations for upright posture, tool use, and brain development.

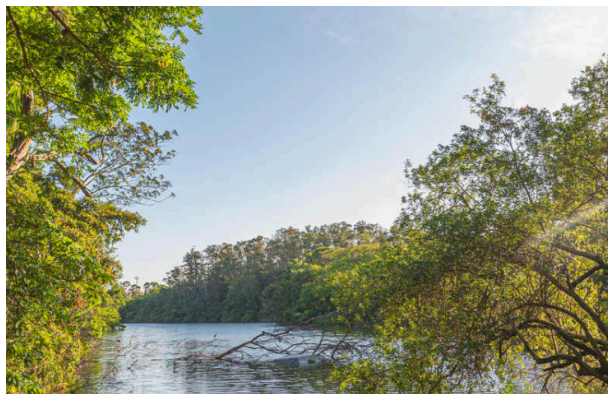


- **Bone stress** increases with body size; large animals have more robust bones and upright posture.



Animal Movement

- **Movement** is based on **contractile proteins** (mainly **actin** and **myosin**).
- **Ameboid movement** uses actin and myosin for pseudopod extension.
- **Cilia** and **flagella** have a 9+2 microtubule arrangement and use **dynein** for movement.



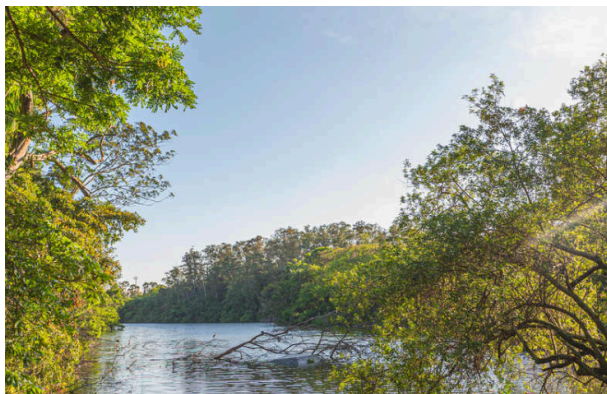


Muscle Types and Structure

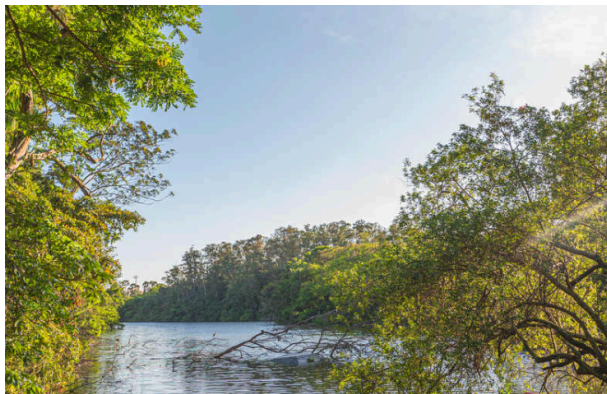
- **Skeletal muscle** is striated, voluntary, and responsible for body movement.
- **Smooth muscle** is non-striated, involuntary, and found in internal organs.
- **Cardiac muscle** is striated, involuntary, and found in the heart.



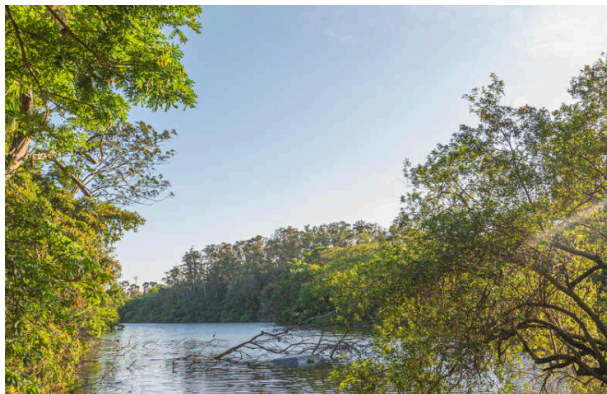
- **Skeletal muscle** is organized into bundles of fibers, each containing myofibrils of actin and myosin.



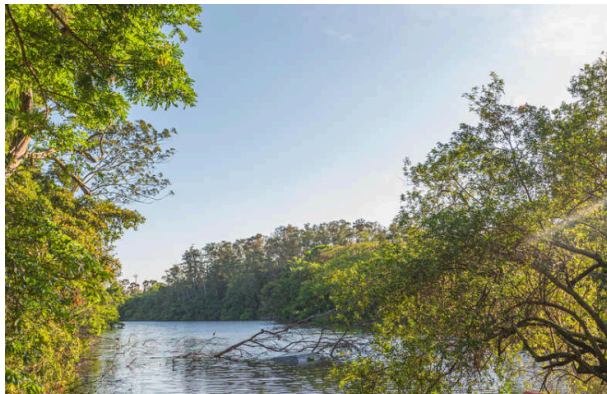
- **Myosin** and **actin** filaments interact for contraction; **troponin** and **tropomyosin** regulate this process.



- **Neuromuscular junctions** transmit nerve signals to muscle fibers, triggering contraction.



- **Excitation-contraction coupling** involves calcium release, troponin activation, and cross-bridge cycling.



Muscle Performance and Energy

- **Muscle contraction** requires ATP, supplied by aerobic metabolism, glycogen, and creatine phosphate.
- The **creatine phosphate** reaction rapidly regenerates ATP during contraction:
 $\text{Creatine-phosphate} + \text{ADP} \rightarrow \text{ATP} + \text{Creatine}$.
- During intense exercise, **anaerobic glycolysis** produces **lactic acid**, leading to **oxygen debt** that must be repaid after exertion.
- **Slow oxidative fibers** are adapted for endurance; **fast glycolytic fibers** for rapid, powerful contractions.
- **Tendons** store and release elastic energy during movement.

